

PROBLEM STATEMENT #42 | Developer Tools & IT Operations

Internal Microservice Catalog & Health Portal

1. Problem Context & Overview

An enterprise developer portal mapping microservice dependencies, aggregating API documentation, and running automated periodic health check pingers with downtime alerts.

Target Stakeholders / Actors: *DevOps Engineer, System Architect*

2. Sample Functional Requirement (FR) Guideline

Sample Requirement: FR-001 [Priority: High]

Description: The system shall ping microservice health endpoints at 30-second intervals and compute rolling 24-hour service availability percentages.

Sample Acceptance Criteria: Pass: Downtime recorded and alert dispatched on 3 consecutive failed health probes;

Fail: Unreachable service marked healthy.

3. Sample Non-Functional Requirement (NFR) Guideline

Sample Requirement: NFR-001 [Type: Performance & Security]

Description: The service catalog dependency graph viewer must render interactive node architectures containing up to 200 services smoothly.

Sample Acceptance Criteria: Pass: Benchmarking tests confirm target latency and security standards under simulated peak load.

Deliverables to Submit: [All through a GitHub Repo in your account]

- **Complete Requirements Table:**

- >> Exactly 5 Functional Requirements (FR-001 to FR-005)
- >> 2 Non-Functional Requirements (NFR-001 & NFR-002) with ID, Type, Description, Priority, Acceptance Criteria, and Rationale.

- **UML Use-Case Diagram:**

- >> Model all actors, primary use cases, and include at least one «include» and one «extend» relationship.

- **Use-Case Flow Specification:**

- >> 1-page document for one core use case detailing Preconditions, Postconditions, Main Success Scenario
- >> One Alternate Flow.